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WO-500-03 — Replace spool SP03 of 1"-P300-HS-5013

Area	A500 Pipe rack	**Task**	SR Spool replacement
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	rack-bent:720 — Pipe rack bent at E 60'-0" — E 60'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	1"-P300-HS-5013
Isometrics	ISO-1"-P300-HS-5013 (generated); owner's drawings ISO-HS-5013		
Permits	General work permit, Hot work permit		

Reason

Steam hose-station drop corroded at the elbow.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 1"-P300-HS-5013 between the isolations; confirm gas-free for hot work.
2. Remove spool 1"-P300-HS-5013-SP03 (5 items, 33'-1-3/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-1"-P300-HS-5013 with the same items (parts list), welding to WS-01; 4 shop welds.
4. Install the spool; make 2 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 420 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 1"-P300-HS-5013/p8, 1"-P300-HS-5013/el9, 1"-P300-HS-5013/p10, 1"-P300-HS-5013/el11, 1"-P300-HS-5013/p12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-HS-5013	1"-P300-HS-5013	1"-P300-HS-5013	1	agrees with the model

ISO-HS-5013 (PDF): ../isometrics-imported/ISO-HS-5013.pdf — the generated isometrics follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:720	structure	Pipe rack bent at E 60'-0"		E 60'-0" N 0.000" EL 122'-0"
1"-P300-HS-5013/p8	item	PIPE 1" P300 BEVELLED ENDS	1"-P300-HS-5013	E 65'-6" N -12" EL 125'-6-11/16"
1"-P300-HS-5013/el9	item	ELBOW 90° R = 5/3 OD 1" P300	1"-P300-HS-5013	E 65'-6" N -11-3/16" EL 125'-11-3/16"
1"-P300-HS-5013/p10	item	PIPE 1" P300 BEVELLED ENDS	1"-P300-HS-5013	E 65'-6" N 5'-0" EL 126'-0"
1"-P300-HS-5013/el11	item	ELBOW 90° R = 5/3 OD 1" P300	1"-P300-HS-5013	E 65'-6" N 10'-11-3/16" EL 125'-11-3/16"
1"-P300-HS-5013/p12	item	PIPE 1" P300 BEVELLED ENDS	1"-P300-HS-5013	E 65'-6" N 11'-0" EL 115'-5-3/4"
1"-P300-HS-5013/A3	item	VALVE, BALL, FLOATING 1" P300	1"-P300-HS-5013	E 65'-6" N -12" EL 124'-4"
1"-P300-HS-5013/A6	item	FLANGE WELD-NECK 1" P300 RAISED FACE	1"-P300-HS-5013	E 65'-6" N -12" EL 125'-15/16"
1"-P300-HS-5013/B3	item	VALVE, BALL, FLOATING 1" P300	1"-P300-HS-5013	E 65'-6" N 11'-0" EL 104'-2-1/8"
1"-P300-HS-5013/B6	item	FLANGE WELD-NECK 1" P300 RAISED FACE	1"-P300-HS-5013	E 65'-6" N 11'-0" EL 104'-11-1/16"
1"-P300-HS-5013/W2	weld	Butt weld 1" P300 (field)	1"-P300-HS-5013	E 65'-6" N -12" EL 125'-3"
1"-P300-HS-5013/W3	weld	Butt weld 1" P300 (shop)	1"-P300-HS-5013	E 65'-6" N -12" EL 125'-10-5/16"
1"-P300-HS-5013/W4	weld	Butt weld 1" P300 (shop)	1"-P300-HS-5013	E 65'-6" N -10-5/16" EL 126'-0"
1"-P300-HS-5013/W5	weld	Butt weld 1" P300 (shop)	1"-P300-HS-5013	E 65'-6" N 10'-10-5/16" EL 126'-0"
1"-P300-HS-5013/W6	weld	Butt weld 1" P300 (shop)	1"-P300-HS-5013	E 65'-6" N 11'-0" EL 125'-10-5/16"
1"-P300-HS-5013/W7	weld	Butt weld 1" P300 (field)	1"-P300-HS-5013	E 65'-6" N 11'-0" EL 105'-1-1/8"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	1"-P300-HS-5013/A3	1"-P300-HS-5013	Close and lock VALVE, BALL, FLOATING 1" P300 (1"-P300-HS-5013/A3)	E 65'-6" N -12" EL 124'-4"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
2	blind	1"-P300-HS-5013/A6	1"-P300-HS-5013	Fit a line blind between 1"-P300-HS-5013/A3 and FLANGE WELD-NECK 1" P300 RAISED FACE (1"-P300-HS-5013/A6), work side	E 65'-6" N -12" EL 124'-10-13/16"		
3	valve	1"-P300-HS-5013/B3	1"-P300-HS-5013	Close and lock VALVE, BALL, FLOATING 1" P300 (1"-P300-HS-5013/B3)	E 65'-6" N 11'-0" EL 104'-2-1/8"		
4	blind	1"-P300-HS-5013/B6	1"-P300-HS-5013	Fit a line blind between 1"-P300-HS-5013/B3 and FLANGE WELD-NECK 1" P300 RAISED FACE (1"-P300-HS-5013/B6), work side	E 65'-6" N 11'-0" EL 104'-8-15/16"		

Parts

Description	Qty	Replaces
PIPE 1" P300 BEVELLED ENDS	3	
ELBOW 90° R = 5/3 OD 1" P300	2	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-500-03-W01	1"-P300-HS-5013/W2	ISO-HS-5013 sh 1 W2	1"-P300-HS-5013	1" P300	0.09375 in	field	WS-01	VT + RT	E 65'-6" N -12" EL 125'-3"		
WO-500-03-W02	1"-P300-HS-5013/W3	ISO-HS-5013 sh 1 W3	1"-P300-HS-5013	1" P300	0.09375 in	shop	WS-01	VT + RT	E 65'-6" N -12" EL 125'-10-5/16"		
WO-500-03-W03	1"-P300-HS-5013/W4	ISO-HS-5013 sh 1 W4	1"-P300-HS-5013	1" P300	0.09375 in	shop	WS-01	VT + RT	E 65'-6" N -10-5/16" EL 126'-0"		
WO-500-03-W04	1"-P300-HS-5013/W5	ISO-HS-5013 sh 1 W5	1"-P300-HS-5013	1" P300	0.09375 in	shop	WS-01	VT + RT	E 65'-6" N 10'-10-5/16" EL 126'-0"		
WO-500-03-W05	1"-P300-HS-5013/W6	ISO-HS-5013 sh 1 W6	1"-P300-HS-5013	1" P300	0.09375 in	shop	WS-01	VT + RT	E 65'-6" N 11'-0" EL 125'-10-5/16"		
WO-500-03-W06	1"-P300-HS-5013/W7	ISO-HS-5013 sh 1 W7	1"-P300-HS-5013	1" P300	0.09375 in	field	WS-01	VT + RT	E 65'-6" N 11'-0" EL 105'-1-1/8"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side
Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-500-03-W01, WO-500-03-W02, WO-500-03-W03, WO-500-03-W04, WO-500-03-W05, WO-500-03-W06	SW-NDE-1
RT (UT where RT is impractical)	100 % of new welds (class S300, high consequence)	WO-500-03-W01, WO-500-03-W02, WO-500-03-W03, WO-500-03-W04, WO-500-03-W05, WO-500-03-W06	SW-NDE-2

Test — Hydrostatic test

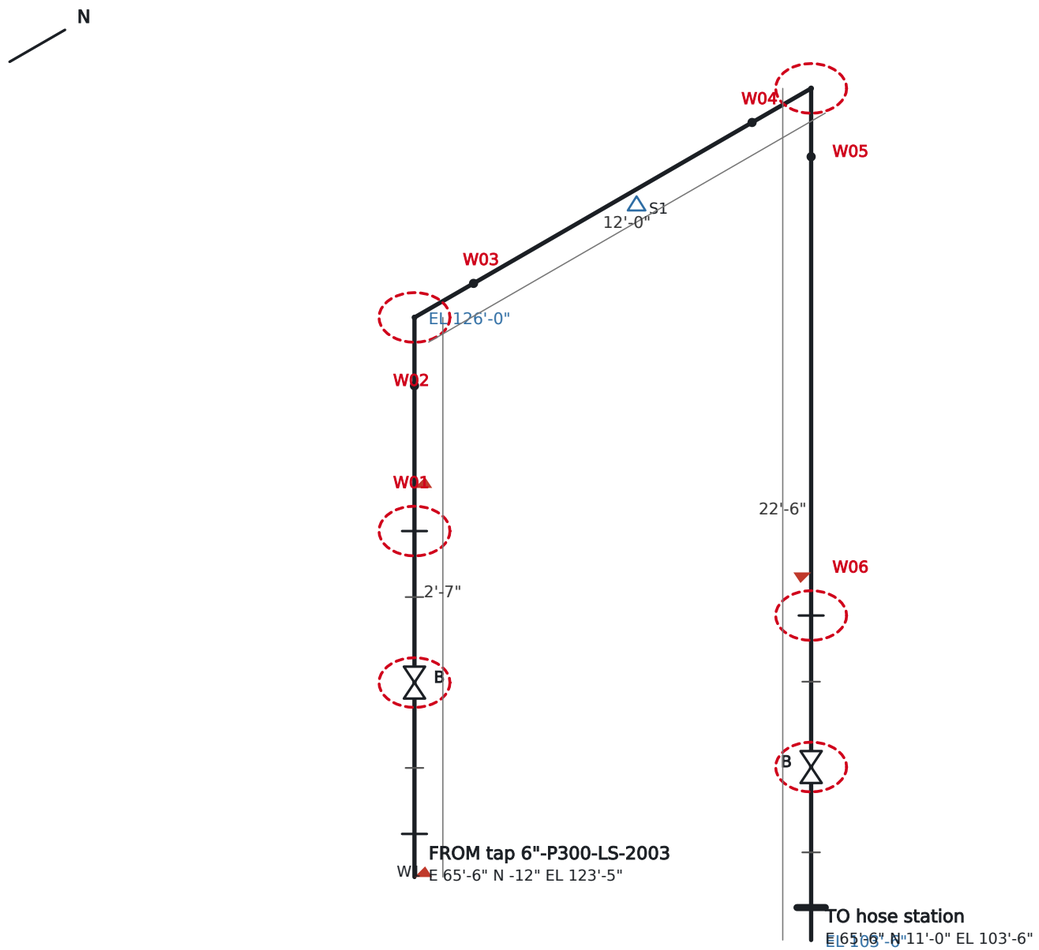
Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	420 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	H	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 420 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 1"-P300-HS-5013 — Replace spool SP03 of 1"-P300-HS-5013



BILL OF MATERIALS

1	FLANGE WELD-NECK 1" P300 RAISED FACE	3
2	GASKET 1" P300 COMPRESSED SHEET	4
3	VALVE, BALL, FLOATING 1" P300	2
4	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P300	4
5	PIPE 1" P300 BEVELLED ENDS	33'-1-3/16"
6	ELBOW 90° R = 5/3 OD 1" P300	2
7	FLANGE BLIND 1" P300 RAISED FACE	1

WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

NEW WELDS (WO-500-03)

WO-500-03-W01 replaces W2
WO-500-03-W02 replaces W3
WO-500-03-W03 replaces W4
WO-500-03-W04 replaces W5
WO-500-03-W05 replaces W6
WO-500-03-W06 replaces W7

1"-P300-HS-5013

CLASS S300 — Low-pressure steam — INSULATED
SERVICE HS · 1" · P300 · DESIGN 300 psi
tap 6"-P300-LS-2003 → hose station
PIPE 33'-1-3/16" · WEIGHT 118 lb

DWG ISO-1"-P300-HS-5013 REV 0 WO-500-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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